

SQL SUBQUERIES & CTE ASSIGNMENT

ROUTE

ASSIGNMENT OVERVIEW

This assignment focuses on using Subqueries and Common Table Expressions (CTEs) to solve SQL problems in a structured and logical way. Instead of writing long and complex queries, you will learn how to break problems into smaller steps and use intermediate results to build clear and readable SQL queries.

OBJECTIVES OF THIS ASSIGNMENT

By completing this assignment, you will be able to:

- Understand how subqueries work and when to use them
- Use subqueries in different SQL clauses such as
- WHERE, SELECT, FROM, and HAVING
- Write queries using CTEs (WITH clause) to organize logic
- Compare subqueries and CTEs and choose the appropriate approach
- Improve query readability and maintainability

Open the provided links and download the dataset files.

https://drive.google.com/file/d/16KdsnhdpD_Q2PSDfO1f27VmqqvePPMTD5/view?usp=drive_link

https://drive.google.com/file/d/1J_0ZZImO18Of3OOWrcBOW3-s7gPBpaiR/view?usp=sharing

<https://drive.google.com/file/d/1ym5cnllYEjw-jMW8ot3Uaa3fEiDR9blv/view?usp=sharing>

<https://drive.google.com/file/d/1kmlj9NmIOkiGbOqGOACtDPXiGxqGDNnv/view?usp=sharing>

Questions:

1. From the following tables, write a SQL query to calculate the average price of each company's products along with the company name.

Return company_name and average_price.

2. Write a SQL query to find all companies where the average item price is greater than the overall average price of all products.

Return company_name and average_price.

3. From the following tables, write a SQL query to find all items whose price is greater than the average price of all items.

Return item_id, item_name, price, and company_id.

4. From the following tables, write a SQL query to find all employees who work in departments whose department budget is greater than the average department budget.

Return emp_id, emp_fname, emp_lname, and dept_code.

5. Write a SQL query to find departments which have more employees than the average number of employees per department.

Return dept_code and dept_name.

6. From the following tables, write a SQL query to find all items whose price is less than the average price of items for the same company.

Return item_id, item_name, price, and company_id.

7. Write a SQL query to find departments that have only one employee assigned to them.

Return dept_code, dept_name, and the count of employees.

8. From the following tables, write a SQL query to find companies whose average item price is equal to the minimum average price among all companies.

Return company_name and average_price.

9. Write a SQL query to find employees whose department budget is less than the average department budget.

Return emp_id, emp_fname, emp_lname, and dept_code.

Open the provided links and download the dataset files.

https://drive.google.com/file/d/16KdsnhdpD_Q2PSDfO1f27VmqqvePPMTD5/view?usp=drive_link

https://drive.google.com/file/d/1J_0ZZImO18Of3OOWrcBOW3-s7gPBpaiR/view?usp=sharing

<https://drive.google.com/file/d/1ym5cnllYEjw-jMW8ot3Uaa3fEiDR9blv/view?usp=sharing>

<https://drive.google.com/file/d/1kmlj9NmIOkiGbOqGOACtDPXiGxqGDNnv/view?usp=sharing>

Questions:

Q1

Create a CTE that calculates the average item price,
then display all items whose price is greater than the average.

Q2

Create a CTE that calculates the total number of items per company,
then display company name and item count.

Q3

Using a CTE, calculate the average price per company,
then display companies where the average price is greater than 2000.

Q4

Create a CTE that joins employees_new and departments_new
to display employee full name and department name.

Q5

Using a CTE, find departments where the department budget is greater than the average budget.

Q6

Create a CTE that calculates the number of employees per department,
then display departments that have more than one employee.

Q7

Create a CTE that calculates the average item price across all companies,
then display companies whose average item price is below the overall average.